

Day 1 – Introduction to python and learn basic syntax

1. Print “Hello World”
2. Swap two variables
3. Find data type of variable
4. Check if a number is Armstrong
5. Find GCD & LCM of two numbers
6. Convert decimal to binary
7. Convert binary to decimal
8. Count frequency of digits in a number
9. Find largest & smallest digit in a number

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1. Print(“Hello World”)

2. Swap Integer

2.1 a = int(input("Enter first number: "))

b = int(input("Enter second number: "))

a, b = b, a

print("After swapping:")

print("a =", a)

print("b =", b)

2.2 a = int(input("Enter first number: "))

b = int(input("Enter second number: "))

temp = a

a = b

b = temp

2.3 a = 5

b = 10

a = a * b

b = a // b

a = a // b

3. Type of value

a = input("Enter a value: ")

print(type(a))

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4.Armstrong number

```
num = int(input("Enter a number: "))
temp = num
sum = 0
digits = len(str(num))
while temp > 0:
    digit = temp % 10
    sum += digit ** digits
    temp //= 10
if sum == num:
    print("Armstrong Number")
else:
    print("Not Armstrong Number")
```

5.GCD(Greatest Common Divisor) & LCM(Least Common Multiply)

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
small = a
if b < a:
    small = b
gcd = 1
for i in range(1, small + 1):
    if a % i == 0 and b % i == 0:
        gcd = i
lcm = (a * b) // gcd
print("GCD =", gcd)
print("LCM =", lcm)
```

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6.Decimal to Binary

```
num = int(input("Enter a decimal number: "))
binary = ""
while num > 0:
    rem = num % 2
    binary = str(rem) + binary
    num = num // 2
print("Binary =", binary)
```

7.Convert binary to decimal

```
binary = input("Enter a binary number: ")
decimal = 0
power = 0
for digit in binary[::-1]:
    decimal += int(digit) * (2 ** power)
    power += 1
print("Decimal =", decimal)
```

8.Count the frequency

```
8.1 num = input("Enter a number: ")
for digit in "0123456789":
    count = num.count(digit)
    if count > 0:
        print(digit, "=", count)
```

```
8.2 num = input("Enter a number: ")
visited = ""
for digit in num:
    if digit not in visited:
        count = 0
```

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```
for ch in num:
    if ch == digit:
        count += 1
print(digit, "=", count)
visited += digit
```

9.Largest and Smallest Digit

```
num = 132
largest = -1
smallest = 10
while num > 0:
    digit = num % 10
    if digit > largest:
        largest = digit
    if digit < smallest:
        smallest = digit
    num = num // 10
print("Largest Digit =", largest)
print("Smallest Digit =", smallest)
```